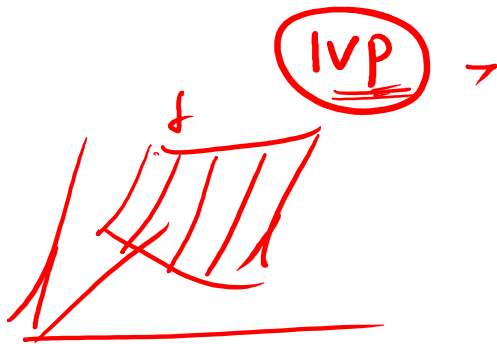


Volatility Models

(4. Surface Calibration - SVI)



Overview



A volatility smile/surface parameterization should be:

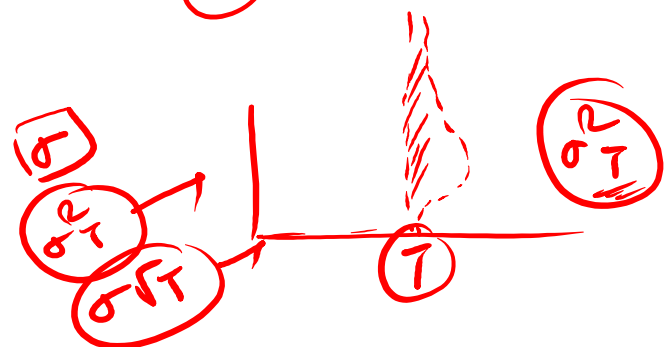
⑤

- ▶ flexible enough to fit market implied vols, ✓
- ▶ simple enough to calibrate quickly, ✓
- ▶ and constrained enough to avoid static arbitrage.

} calendar spread
butterfly }

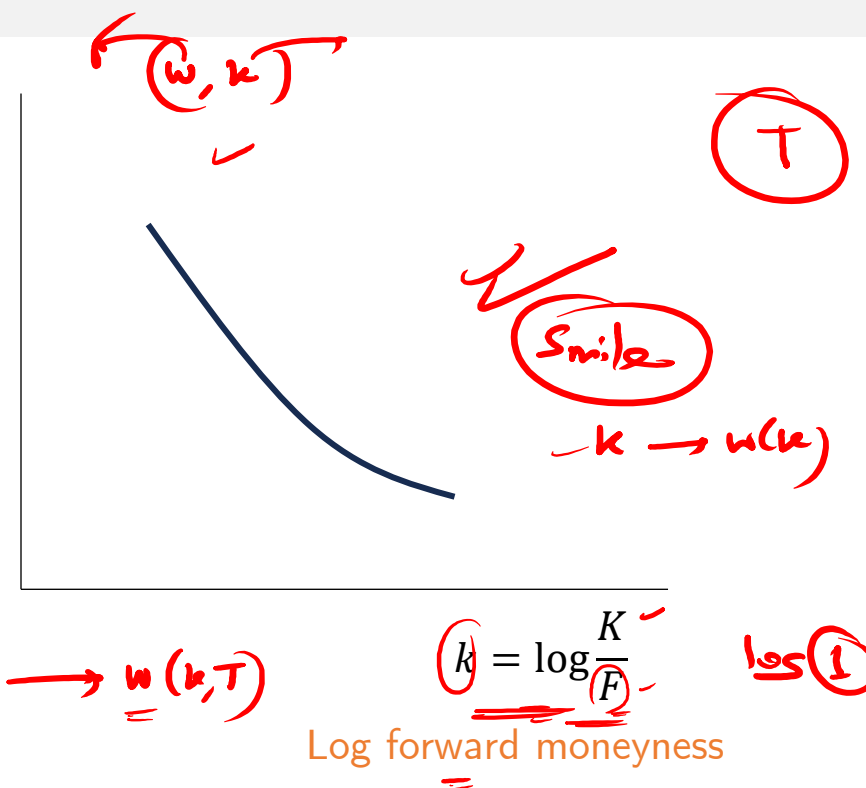
Highlights

- ▶ how to characterize static arbitrage in implied volatility surfaces,
- x ▶ how standard **SVI** slices can still be arbitrageable,
- ▶ and how the **SSVI** surface class gives a simple closed-form surface that can be made arbitrage-free under explicit conditions.



$$w(k, T) = \sigma_{BS}^2(k, T)T$$

Total implied variance



Define implied variance = $v(k, T) = \sigma_{BS}^2(k, T) = w(k, T)/T$

A slice is $k \rightarrow w(k, T)$ for a fixed maturity T

A surface is $(k, T) \rightarrow w(k, T)$

Static Arbitrage

A volatility surface is free of static arbitrage iff:

- ▶ it has no calendar spread arbitrage

→ $\frac{\partial w(k,T)}{\partial T} \geq 0$ for all k, T (Total variance must be non-decreasing in maturity for fixed moneyness)

- ▶ every maturity slice has no butterfly arbitrage

For a fixed slice $w(k)$, the risk-neutral density should be non-negative.

A slice is free of butterfly arbitrage iff

(i) $g(k) = \left(1 - \frac{k w'(k)}{2 w(k)}\right)^2 - \frac{w'(k)^2}{4} \left(\frac{1}{w(k)} + \frac{1}{4}\right) + \frac{w''(k)}{2} \geq 0$ for all k and

(ii) $\lim_{k \rightarrow +\infty} d_+(k) = -\infty$ (means call option price should approach 0 when k tends to infinity)

$S N(d_1) - K e^{-rT} N(d_2)$ $d_1 = d_+$, $d_2 = d_-$

$w(k,T) = \sigma_{BS}^2(k,T) \cdot T$



$\sigma_{BS}^2(k, T_2) T_2 \geq \sigma_{BS}^2(k, T_1) T_1$
if $T_2 > T_1$

$\left(\frac{\partial^2 w}{\partial k^2} \right) > 0$



$\frac{\partial w}{\partial k}$, $\frac{\partial^2 w}{\partial k^2}$

$k = \log\left(\frac{K}{F}\right)$

Proof of “butterfly arbitrage” condition

Write BSM call option price as

$$C_{BS}(k, w(k)) = S \left(N(d_+(k)) - e^{-k} N(d_-(k)) \right)$$

where $d_{\pm}(k) = -k/\sqrt{w(k)} \pm \sqrt{w(k)}/2$

$$\ln\left(\frac{S}{k}\right) + \left(r + 0.5\sigma^2\right)T$$

$$\begin{aligned} S \cdot e^{-k} &= S \cdot e^{-\ln\left(\frac{k}{F}\right)} \\ &= S \cdot \frac{F}{k} = \frac{S \cdot k}{S e^{rT}} = \frac{k e^{-rT}}{1} \end{aligned}$$

the probability density function p may be computed from the call price function C as

$$p(k) = \frac{\partial^2 C}{\partial K^2} \bigg|_{K=Fe^k} = \frac{\partial^2 C_{BS}(k, w(k))}{\partial K^2} \bigg|_{K=Fe^k}$$

Differentiation leads to

$$p(k) = \frac{g(k)}{\sqrt{2\pi w(k)}} \exp\left(-\frac{1}{2} d_-(k)^2\right)$$

Raw SVI

There are 3 versions of **SVI** (Stochastic volatility inspired) formulations. We can move back and forth between the three types of parametrizations. The one below is called the **Raw SVI**

with constraints:

- $b \geq 0$ ✓
- $|\rho| < 1$ ✓
- $\sigma > 0$ ✓
- and positivity condition on the minimum variance ; $a + b\sigma\sqrt{1 - \rho^2} \geq 0$

Interpretation of raw parameters

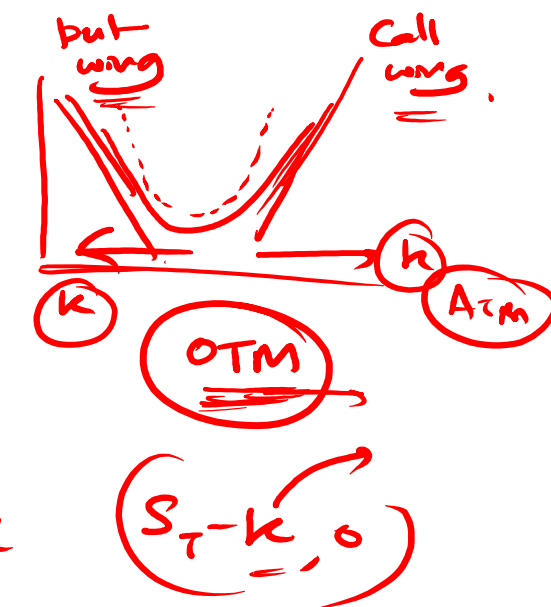
- a : increasing ' a ' increases the level of variance (vertical translation of the smile)
- b : increasing ' b ' increases the slope of calls/puts in the wings (tightens the smile)
- ρ : increasing ρ leads to counter-clockwise rotation of the smile
- m : increasing ' m ' shifts the smile to the right
- σ : increasing ' σ ' reduces the ATM curvature of the smile

$$w(k) = a + b \left[\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2} \right]$$

$$\frac{\partial w}{\partial k} = 0$$

$$w^*_{\min} > 0$$

$$k=0$$



Natural SVI

The natural SVI parametrization goes like this

$$w(k) = \Delta + \frac{\omega}{2} \left[1 + \zeta \rho (k - \mu) + \sqrt{(\zeta (k - \mu) + \rho)^2 + (1 - \rho^2)} \right]$$

where $\omega \geq 0, \Delta \in R, \mu \in R, |\rho| < 1$ and $\zeta > 0$

The mapping between Raw and Natural parametrization is given as

$$\underline{\text{Raw}} = \left(\Delta + \frac{\omega}{2} (1 - \rho^2), \frac{\omega \zeta}{2}, \rho, \mu - \frac{\rho}{\zeta}, \frac{\sqrt{1 - \rho^2}}{\zeta} \right)$$

The SVI Jump wings SVI-JW

Neither the Raw nor the Natural SVI are intuitive to traders. We define the SVI-JW parameters for a given maturity slice T as $\{v_T, \psi_T, p_T, c_T, \tilde{v}_T\}$ from the raw SVI parameters as follows

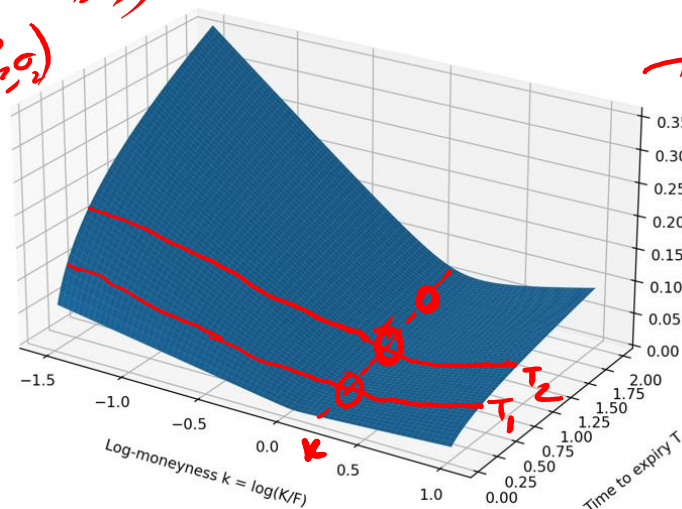
$$\begin{aligned}
 \textcircled{T} \quad v_T &= \frac{a + b\{-\rho m + \sqrt{m^2 + \sigma^2}\}}{T} & \psi_T &= \frac{1}{\sqrt{w_T}} \frac{b}{2} \left(-\frac{m}{\sqrt{m^2 + \sigma^2}} + \rho \right) & p_T &= \frac{1}{\sqrt{w_T}} b(1 - \rho) \\
 c_T &= \frac{1}{\sqrt{w_T}} b(1 + \rho) & \tilde{v}_T &= \frac{a + b\sigma\sqrt{1 - \rho^2}}{\textcircled{T}} \quad \} > 0
 \end{aligned}$$

- $\textcircled{v_T}$ gives the ATM variance;
- $\psi_T = \left. \frac{\partial \sigma_{BS}(k, T)}{\partial k} \right|_{k=0}$ gives the ATM skew;
- p_T gives the slope of the left (put) wing;
- c_T gives the slope of the right (call) wing;
- $\tilde{v}_T$ is the minimum implied variance.

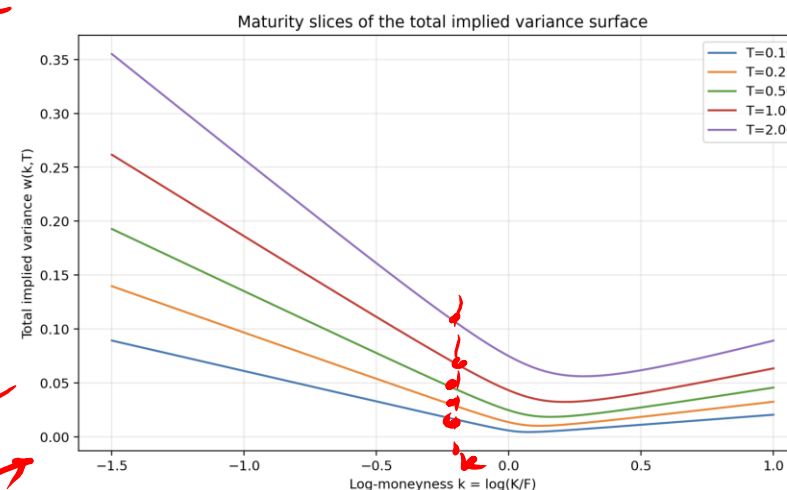
SVI limitation - I

Checking calendar arbitrage for raw SVI slices involves checking whether slices intersect across maturities;

$(T_1) \rightarrow (a_1, b_1, m_1, \rho_1, \sigma_1)$
 $(T_2) \rightarrow (a_2, b_2, m_2, \rho_2, \sigma_2)$



$w(k, T)$



Slices should not intersect

$$a_1 + b_1 \left[\rho_1(k - m_1) + \sqrt{(k - m_1)^2 + \sigma_1^2} \right] \neq a_2 + b_2 \left[\rho_2(k - m_2) + \sqrt{(k - m_2)^2 + \sigma_2^2} \right] \quad \text{for } k \in \mathbb{R}$$

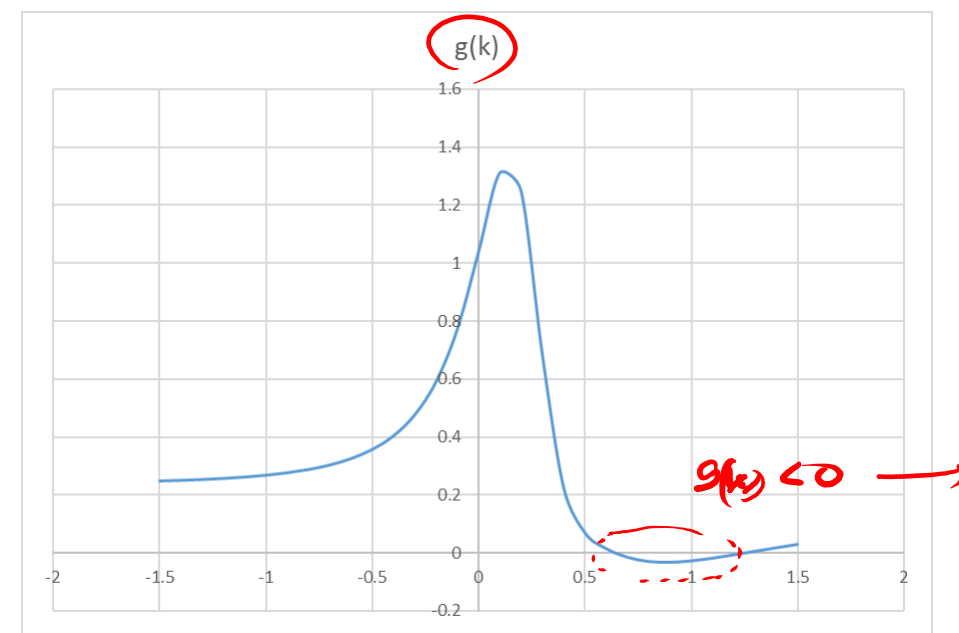
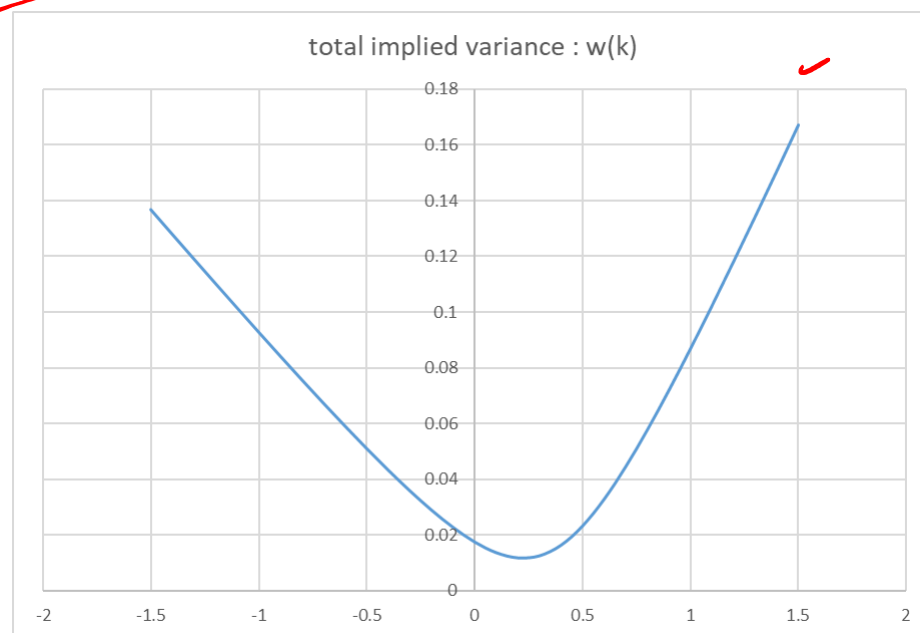
SVI limitation - II

Butterfly arbitrage conditions are much harder to express directly in raw SVI parameters. Here is one set of parameters that lead to violation of butterfly arbitrage. Note $g(k) < 0$

Handwritten notes:

- $g(k) \geq 0$
- $\sqrt{a+b} \geq 2m\rho\sigma$

Parameters: $(a, b, m, \rho, \sigma) = (-0.0410, 0.1331, 0.3586, 0.3060, 0.4153)$



Surface SVI : SSVI

We now introduce SSVI (Surface SVI) which is an extension of natural parametrization for any maturity $T \geq 0$. Define at-the-money (ATM) implied total variance $\theta_T = \sigma_{BS}^2(0, T)T$. Let $\varphi(\theta_T)$ be a smooth function.

Then SSVI is given as

$\theta_T \rightarrow$ ATM total implied variance

$$w(k, \theta_T) = \frac{\theta_T}{2} \left[1 + \rho \varphi(\theta_T) k + \sqrt{(\varphi(\theta_T) k + \rho)^2 + (1 - \rho^2)} \right]$$

It's just the natural SVI form with

$$\Delta = 0, \mu = 0, \omega = \theta_T, \zeta = \varphi(\theta_T).$$

So SSVI is basically natural SVI with maturity-dependent structural parameters tied to ATM total variance.

Arbitrage conditions for SSVI

SSVI is free of calendar arbitrage iff

$$\frac{\partial \theta_T}{\partial T} \geq 0$$

skew



$$0 \leq \frac{\partial(\theta \varphi(\theta))}{\partial \theta} \leq \frac{1 + \sqrt{1 - \rho^2}}{\rho^2} \varphi(\theta)$$

ATM total variance increasing with maturity, and
the skew term structure not moving "too aggressively."

SSVI is free of butterfly arbitrage iff

A sufficient set of conditions is:

$$\theta \varphi(\theta)(1 + |\rho|) < 4 \quad \text{and}$$

$$\theta \varphi(\theta)^2(1 + |\rho|) \leq 4$$

for all relevant $\theta > 0$.

SSVI and SVI-JW link



The SSVI parameters are linked to SVI-JW

$$\underline{v_T} = \frac{\theta_T}{T}$$

$$\underline{\psi_T} = \frac{\rho}{2} \sqrt{\theta_T} \varphi(\theta_T)$$

$$\underline{p_T} = \frac{1}{2} \sqrt{\theta_T} \varphi(\theta_T) (1 - \rho)$$

$$\underline{c_T} = \frac{1}{2} \sqrt{\theta_T} \varphi(\theta_T) (1 + \rho)$$

$$\underline{\tilde{v}_T} = \frac{\theta_T}{T} (1 - \rho^2)$$

Two useful choices of $\varphi(\theta_T)$

Heston-like choice

$$\varphi(\theta) = \frac{1}{\lambda\theta} \left(1 - \frac{1 - e^{-\lambda\theta}}{\lambda\theta} \right)$$

This is motivated by Heston-style skew decay and satisfies the calendar conditions under the theorem.

Power-law choice

$$\varphi(\theta) = \eta\theta^{-\gamma}, \eta > 0, 0 < \gamma < 1$$

The paper then recommends a modified version for large maturities:

$$\varphi(\theta) = \frac{\eta}{\theta^\gamma(1+\theta)^{1-\gamma}}$$

which can give an entire surface free of static arbitrage provided $\eta(1+|\rho|) \leq 2$